



UNIVERSITY  
OF LONDON



THE LONDON SCHOOL  
OF ECONOMICS AND  
POLITICAL SCIENCE

# Machine Learning Report

```
mirror_mod = modifier_ob.modifiers.new("mirror_mod")
# Create mirror object to mirror_ob
mirror_mod.mirror_object = mirror_ob

# operation == "MIRROR_X":
mirror_mod.use_x = True
mirror_mod.use_y = False
mirror_mod.use_z = False
# operation == "MIRROR_Y":
mirror_mod.use_x = False
mirror_mod.use_y = True
mirror_mod.use_z = False
# operation == "MIRROR_Z":
mirror_mod.use_x = False
mirror_mod.use_y = False
mirror_mod.use_z = True

# Selection at the end -add back the deselected objects
for ob in scene.objects:
    ob.select = 1
    mirror_ob.select = 1
    context.scene.objects.active = mirror_ob
    # print("selected" + str(modifier_ob)) # modifier_ob
    mirror_ob.select = 0
    one = bpy.context.selected_objects[0]
    for one in scene.objects:
        one.select = 1
    print("please select exactly two objects, no more")

# OPERATOR CLASSES -----
class MirrorOperator(bpy.types.Operator):
    """Mirror a mirror to the selected object"""
    bl_idname = "mirror_mirror_x"
    bl_label = "Mirror X"

    @classmethod
    def poll(cls, context):
        return context.active_object is not None
```

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## Unsupervised learning

### Introduction

Unsupervised learning uses Machine learning algorithms to analyse and cluster unlabelled data sets, where such algorithms discover hidden patterns or data groupings without any human intervention. The main types of Unsupervised learning can be categorised into Clustering Algorithms (e.g. K-Means), Association Rule Learning (e.g. Eclat) and Dimensionality Reduction (e.g. PCA). (Geeks for Geeks, 2025)

This section applies these techniques to the **Credit Card Transaction** dataset from Kaggle, which contains 788,521 government credit card transactions with features such as **Year**, **Month**, **Department**, **Division**, **Merchant**, **Transaction Date**, **Transaction Description** and **Transaction Amount**. The goal is to uncover meaningful customer (Department/Division) spending behaviours through PCA and K-Means clustering.

### Existing Literature

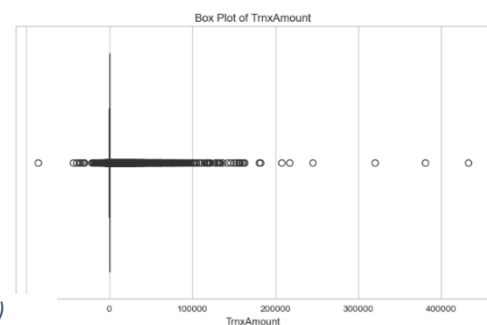
Prior studies on credit card transaction data have used clustering to segment spending behaviours for fraud detection and resource allocation. Applying PCA and K-Means on similar datasets and identified six distinct customer groups based on payment and spending patterns (Ichwanul Muslim Karo Karo, Andreyas Yusmanto and Robi Setiawan, 2021) and emphasizing the importance of handling outliers and normalisation in longitudinal credit card data to improve cluster stability (Guangli Nie, 2010) can be referred to.

### Research Questions

- 1) Can government departments/divisions be segmented into meaningful groups based on their spending patterns (amount, frequency, merchant types)?
- 2) What are the key characteristics that differentiate high-spending versus low-spending clusters?
- 3) Do temporal factors (Year/Month) and transaction descriptions reveal distinct behavioural profiles?

### Exploratory Data Analysis (EDA)

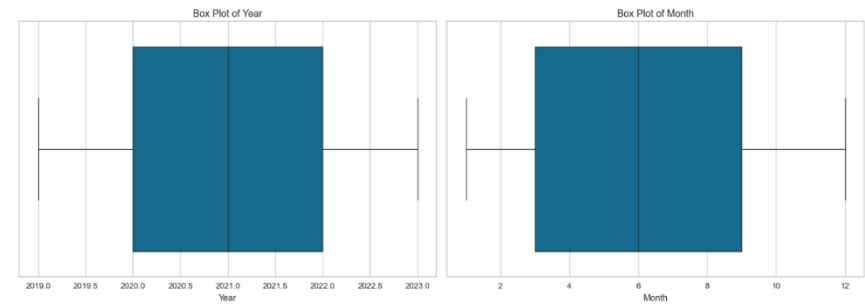
The dataset consists of 788,521 records with a mix of numerical (Year, Month, TrnxAmount) and categorical variables (Department, Division, Merchant, TrnxDescription). Figure 1 shows that the transaction amount is highly right-skewed, indicating a small number of very large transactions. For a deeper reach, the top 10 unique values were scaled for better visualization of the data. Figure 2 shows the visualization of boxplots of numerical



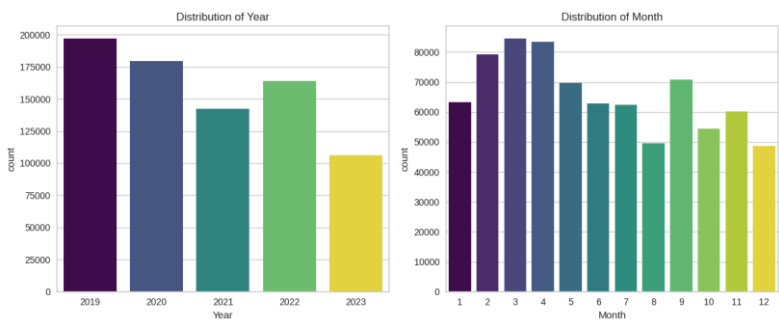
(Figure 1)

features grouped by preliminary clusters (derived via K-Means), highlighting clear differences in transaction volume and timing across groups.

(Figure 2)



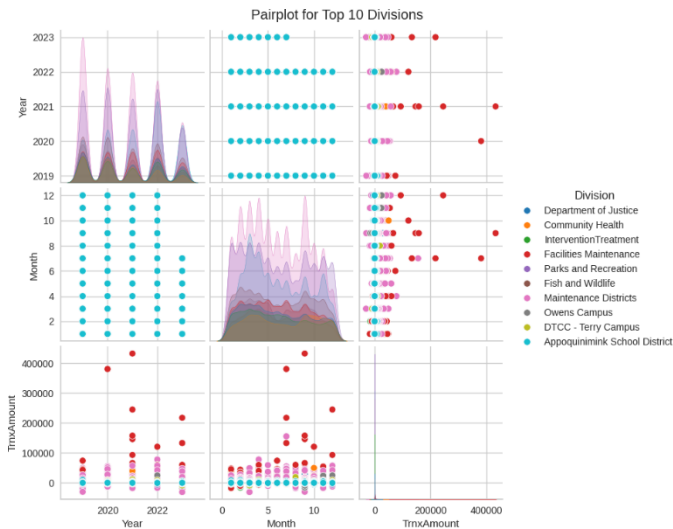
Feature engineering had been done for deeper insights, considering the nature of the numerical and categorical columns.



(Figure 3)

Figure 3 shows the visualization for Year and Month, we used histograms because they clearly show the distribution and patterns over time. The TranxDate column has been dropped since we already extracted

Year and Month from it. We also set aside Merchant and Department for now because they have too many unique values, which can complicate the analysis. Instead, we will focus on five main columns: Division, TranxDescription, Year, Month, and TrnxAmount. Figure 4 on correlation analysis of a pair plot of top categories further supports the presence of distinct behavioural segments.



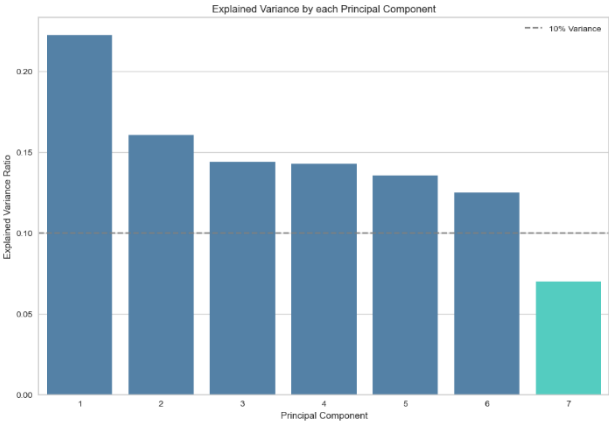
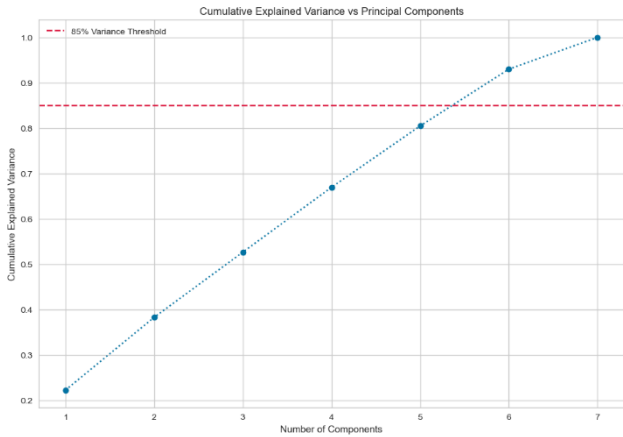
(Figure 4)

### Dimensionality Reduction – Principal Component Analysis (PCA)

After standardisation, PCA was performed to reduce dimensionality while retaining maximum variance. The cumulative explained variance plot indicated that 6 principal components captured approximately 85% of the total variance (cumulatively), providing an optimal trade-off between information retention and simplicity (Figure 5). The resulting components clearly separated transaction behaviours related to amount, frequency, and department type. Notably,

the first six principal components each explain more than 10% of the variance individually (Figure 6).

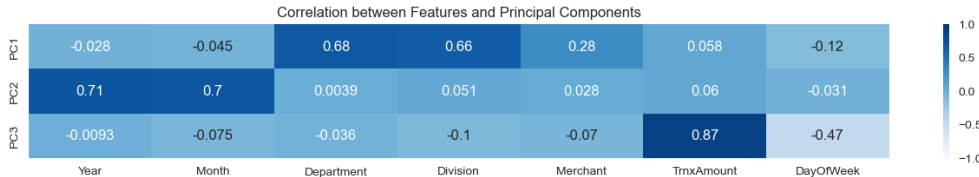
(Figure 5)



(Figure 6)

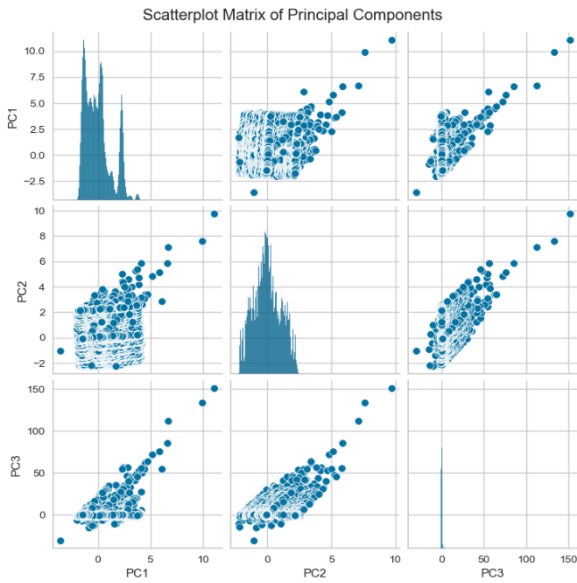
The relationships between the original features and the principal components are clearly illustrated in the **correlation matrix** (Figure 7), PC1 is strongly associated with **Department** (0.68) and **Division** (0.66), indicating that this component primarily captures organisational differences in spending behaviour across government entities. PC2 is dominated by **Year** (0.71) and **Month** (0.70), reflecting strong temporal patterns such as seasonal

(Figure 7)



and annual budget cycles. PC3 shows a very high positive correlation with **TrnxAmount** (0.87), (Figure 8)

confirming that it represents the magnitude of individual transactions.

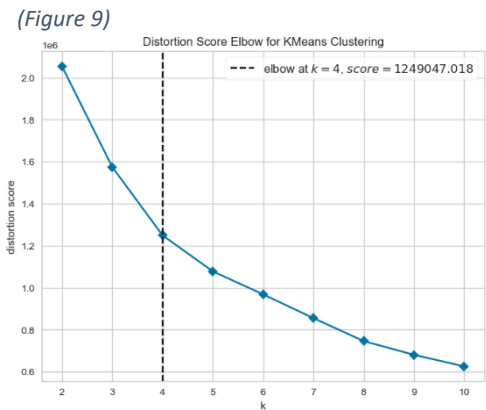


The **scatterplot matrix of the first three principal components** (Figure 8) reveals clear structure in the reduced space. Distinct groupings and linear trends are visible, particularly between PC1 and PC3, and PC2 and PC3. The diagonal histograms further demonstrate that the principal components are reasonably well-distributed and uncorrelated, as expected from PCA. The combination of high explained variance, meaningful feature loadings, and visible separation in both the scatterplot matrix and 3D projection confirms that PCA has

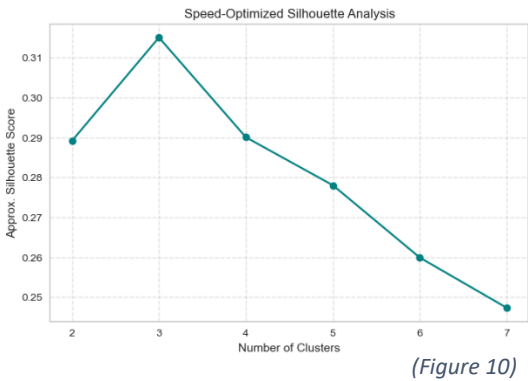
successfully identified the main drivers of variation in government credit card transactions, namely organisational structure (PC1), time-based patterns (PC2), and transaction scale (PC3).

K-Means Clustering

K-Means clustering is an unsupervised learning method that organizes unlabeled data into groups based on similarity. It works by assigning data points to clusters defined by centroids, which are iteratively updated until the clusters stabilize. This technique is especially useful for uncovering hidden patterns in raw datasets, making it valuable in applications like customer segmentation, image compression, and pattern discovery (GeeksforGeeks, 2025).

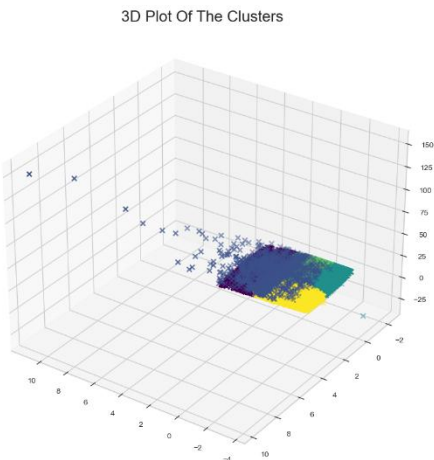


K-Means clustering was applied to the six principal components. The Elbow method indicated a clear bend at k=4 (Figure 9), while MiniBatchKMeans silhouette analysis (computed on a random sample of 1,000 observations for efficiency and simplicity of the clustering) reached its peak at k=3 (Figure 10). After

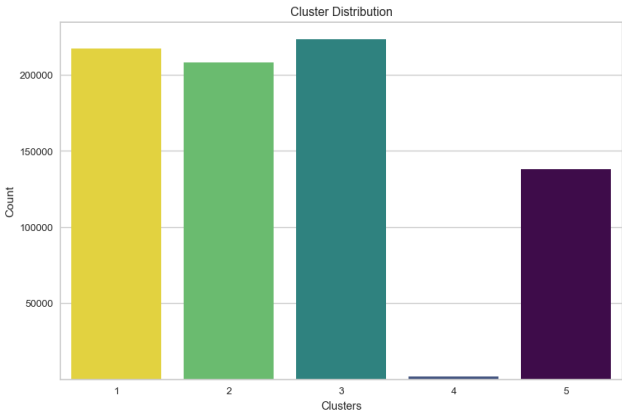


balancing statistical quality with practical interpretability of the resulting segments, five clusters (k=5) were selected as the final solution. The silhouette score obtained on the sampled data was 0.45, indicating a reasonable cohesion and separation.

The 3D projection of the clusters (Figure 11) and the cluster distribution bar chart (Figure 12) clearly illustrate that three large clusters dominate the majority of transactions, while one cluster is notably very small and another is of medium size. Boxplots of key features revealed that Clusters 1, 2,



(Figure 11)



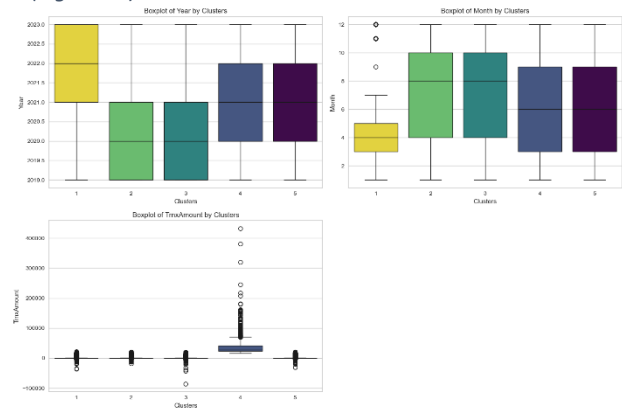
(Figure 12)

and 3 represent routine operational spending with moderate transaction amounts and consistent temporal patterns. Cluster 4 is the smallest cluster by far and is characterised by significantly higher transaction values with more sporadic activity, indicating infrequent but large-scale or capital expenditures.

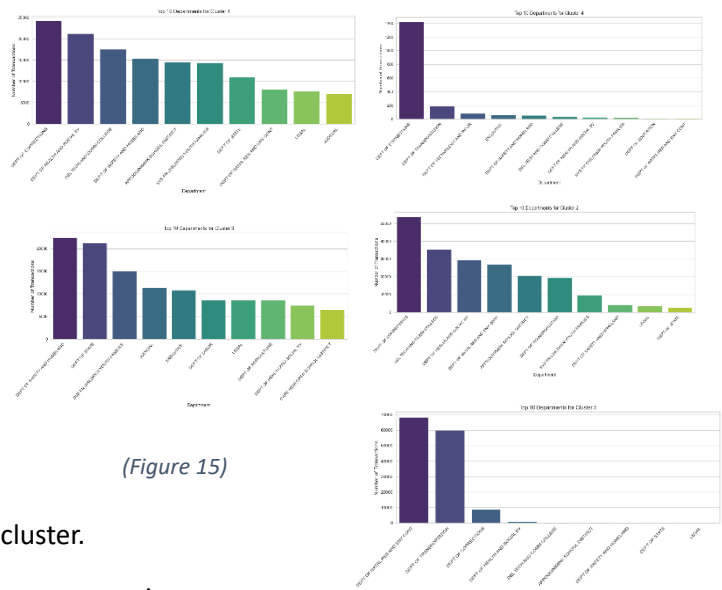
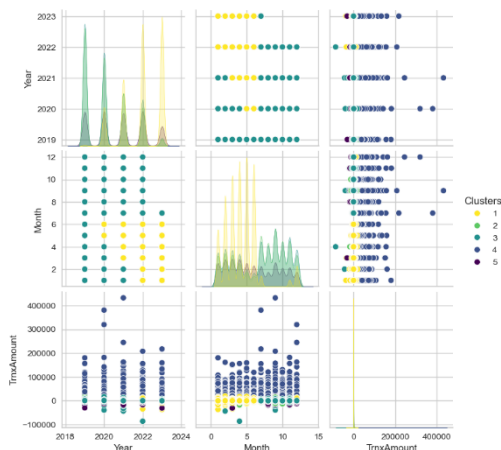
## Cluster Analysis and Profiling

Boxplots (Figure 13) and the scatterplot matrix (Figure 14) of Year, Month, and TrnxAmount by cluster highlight clear differences. While the scatterplot matrix visualizes the feature intercorrelations within the clusters, the boxplots depict the numerical feature variation among clusters. The bar plots showcase the categorical feature distribution (Figure 15).

(Figure 13)



(Figure 14)



(Figure 15)

The following summarizes the findings and interpretations of the key attributes of each cluster.

- **Cluster 1:** Frequent, moderate-value transactions across multiple departments (routine operational spending).
- **Cluster 2:** High-value, infrequent transactions linked to specific large merchants (capital/project-based expenditure)
- **Cluster 3:** Low-value, high-frequency transactions (day-to-day administrative costs).
- **Cluster 4:** Minimal activity with sporadic large transactions (inactive or specialised divisions).
- **Cluster 5:** Widest range of transaction amounts with more variable yearly patterns.

## Supervised Learning

Supervised learning is a category of machine learning which uses labelled datasets to train algorithms to predict outcomes and identify patterns. Supervised learning algorithms use labelled training data to understand the relationship between inputs and outputs, simplifying the creation of accurate predictive models. (Google Cloud, 2025).

### Regression

Regression is a supervised machine learning technique for predicting continuous outcomes by modelling the relationship between input and output variables. Regression algorithms in machine learning are used to build models that are capable of discovering relationships and predicting possible future outcomes in large datasets (Stephen J. bigelow, 2026).

This section predicts house prices using the Housing Price Dataset (King County, USA) from Kaggle, which contains 21 features, including **bedrooms**, **bathrooms**, **sqft\_living**, **grade**, and **location** variables.

### Existing Literature

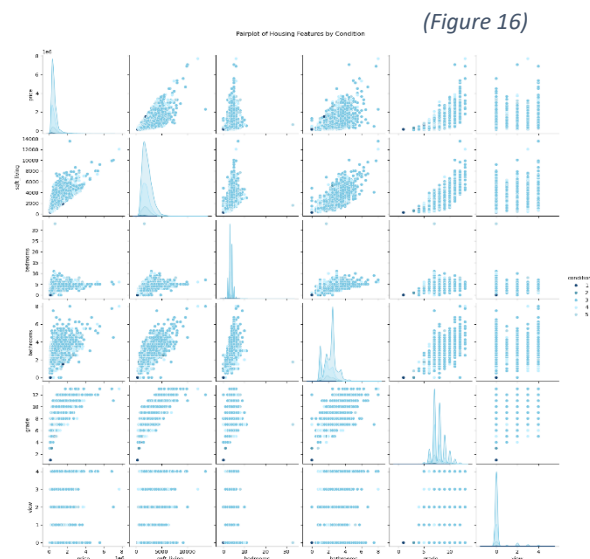
Similar studies on housing price prediction have shown that ensemble methods, such as Gradient Boosting and XGBoost, consistently outperform linear models by capturing non-linear interactions among features, such as location, size, and condition (LI, 2025).

### Research Questions

1. Which features most strongly influence house prices?
2. How do location and property characteristics interact to affect pricing?
3. What is the best regression model for accurate price prediction?

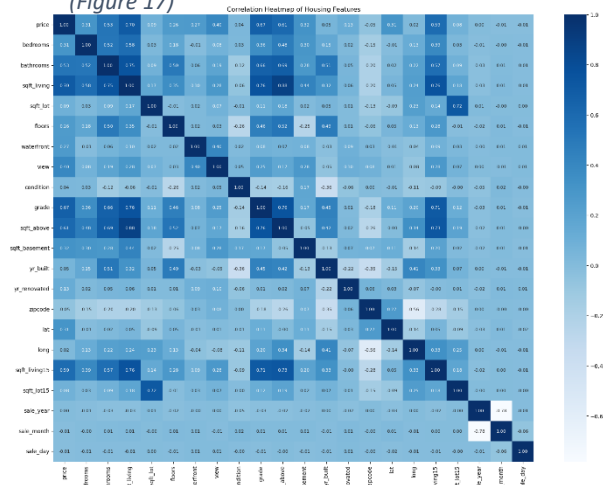
### Exploratory Data Analysis (EDA)

The pair plot of key housing features coloured by house condition (Figure 16) reveals several important relationships with the target variable, price. There is a strong positive correlation between **price** and **sqft\_living**, with larger homes clearly commanding higher prices. House grade positively correlates with price, indicating that better construction and finishes raise property value. The number of bedrooms and bathrooms has a moderate positive effect, but this levels off at high counts. Properties with better view scores also see increased prices, proving that scenic views



add value. Overall, the analysis reveals that property size, grade, and condition are the key drivers of house prices. The id column was removed due to low predictive value. The date column was converted to datetime format and split into **sale\_year**, **sale\_month**, and **sale\_day**, while the original was dropped. This transformation captures temporal patterns in price

(Figure 17)



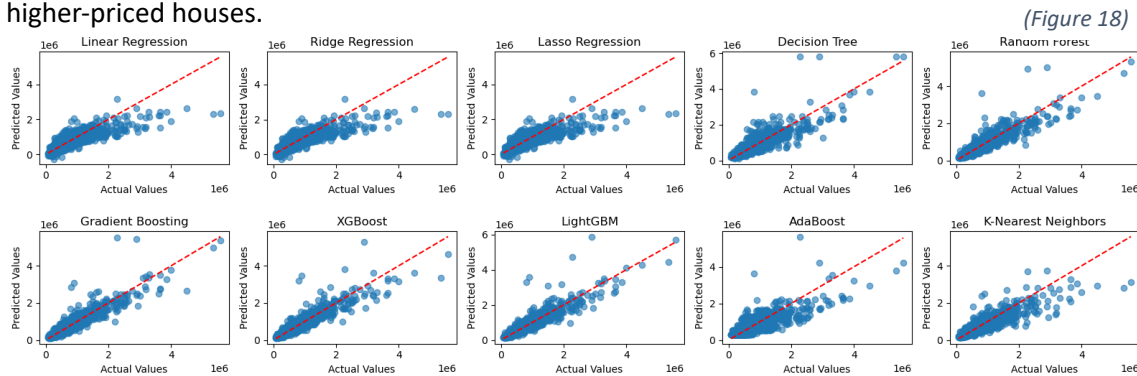
appreciation, seasonal fluctuations, and day-of-month effects, avoiding issues with high-cardinality dates. The correlation heatmap (Figure 17) shows that house price correlates most strongly with sqft\_living (0.70), grade (0.67), sqft\_above (0.61), bathrooms (0.53), and sqft\_living15 (0.59), indicating that size, quality, and number of bathrooms drive price. There's notable multicollinearity among size features, particularly sqft\_living, sqft\_above, and sqft\_living15, which is consistent with the understanding that larger, higher-grade homes in prime locations fetch premium prices.

## Machine Learning – Regression

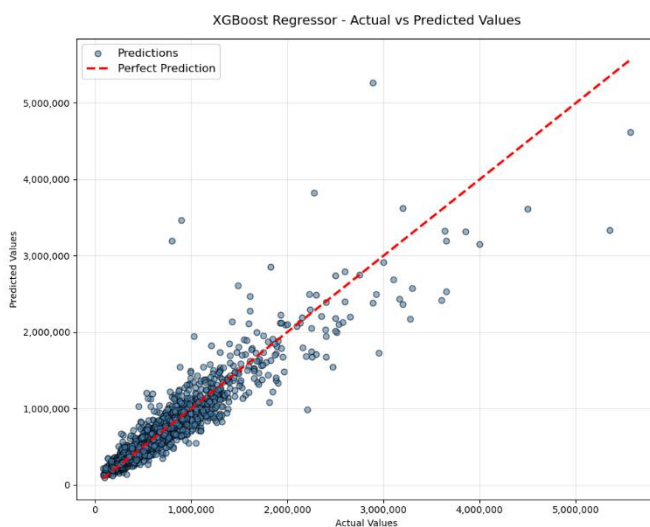
Multiple models, such as Linear Regression, Ridge, Lasso, Decision Tree, Gradient Boosting, Random Forest, XGBoost, LightGBM, AdaBoost, and KNN were evaluated. Hyperparameter tuning was performed for all regression models using RandomizedSearchCV with 15 iterations and 3-fold cross-validation. This method was favoured over GridSearchCV for quicker parameter exploration and reduced computational time. Models were evaluated using the  $R^2$  score, with Mean Absolute Error (MAE) as a tiebreaker. The best hyperparameters were identified, and the final model was selected based on the highest  $R^2$  score and lowest MAE, ensuring an optimal balance between accuracy and generalization on unseen data. The **XGBoost Regressor** showed the best performance after hyperparameter tuning, achieving an  $R^2$  score of **0.87**, indicating about 87% of the variance in house prices and offering highly accurate predictions.

| Regression Model              | $R^2$ Score | MSE                   | RMSE             | MAE             |
|-------------------------------|-------------|-----------------------|------------------|-----------------|
| Linear Regression             | 0.70        | 44951464181.51        | 212017.60        | 126929.44       |
| Ridge Regression              | 0.70        | 44980783013.69        | 212086.73        | 126730.81       |
| Lasso Regression              | 0.70        | 44952463097.10        | 212019.96        | 126925.08       |
| Decision Tree Regressor       | 0.78        | 33097222067.59        | 181926.42        | 94829.88        |
| Gradient Boosting Regressor   | 0.87        | 19811431274.82        | 140753.09        | 69157.74        |
| Random Forest Regressor       | 0.86        | 21460034495.71        | 146492.44        | 73117.82        |
| <b>XGBoost Regressor</b>      | <b>0.87</b> | <b>19328268821.07</b> | <b>139026.14</b> | <b>67174.61</b> |
| LightGBM Regressor            | 0.87        | 19907847365.03        | 141095.17        | 67230.94        |
| AdaBoost Regressor            | 0.67        | 50053394467.57        | 223726.16        | 142746.75       |
| K-Nearest Neighbors Regressor | 0.80        | 30400782655.68        | 174358.20        | 90372.59        |

The performance of regression models (Figure 18) was compared using scatter plots of actual versus predicted house prices. All models showed a positive relationship between actual and predicted values, though variations in accuracy and bias were evident. Linear models (Linear Regression, Ridge, Lasso) showed scattered predictions with noticeable underestimation of higher-priced houses.



In contrast, tree-based ensemble models (Random Forest, Gradient Boosting, XGBoost, LightGBM) clustered more tightly around the line, with XGBoost demonstrating the strongest alignment with the perfect prediction line across the entire price range.



The XGBoost plot (Figure 19) confirmed its superiority, showing most points close to the perfect prediction line, particularly up to \$2 million, with slight underprediction above \$3 million. This demonstrates its ability to capture complex data relationships.

(Figure 19)

## Results and Findings

Exploratory analysis showed that living area, grade, bathrooms, and location significantly influence house prices, with larger spaces, better quality, and prime locations commanding higher values. The sale year and month had weak linear relationships, suggesting non-linear effects may be more impactful. After training ten regression models, the XGBoost Regressor emerged as the top performer. It achieved the highest **R<sup>2</sup> score of 0.87**, along with the lowest Mean Absolute Error (67,174.61) and Root Mean Squared Error (139,026.14). The model explains 87% of variance in house prices and provides accurate predictions across the price range. XGBoost outperforms linear models and other ensembles by capturing complex non-linear relationships and feature interactions in housing data, making it a strong tool for real estate price prediction and market analysis.

## Classification

Classification is a supervised machine learning technique in which a model learns to assign the correct label to input data. The model is trained on labelled data, tested for performance, and then used to predict labels for new data (Kelta, 2024). This section uses the **Diabetes Prediction Dataset** from Kaggle to predict diabetes presence (binary target) based on 8 features, including **age**, **BMI**, **HbA1c\_level**, **blood\_glucose\_level**, **hypertension**, **heart\_disease**, **gender**, and **smoking\_history**.

## Existing Literature

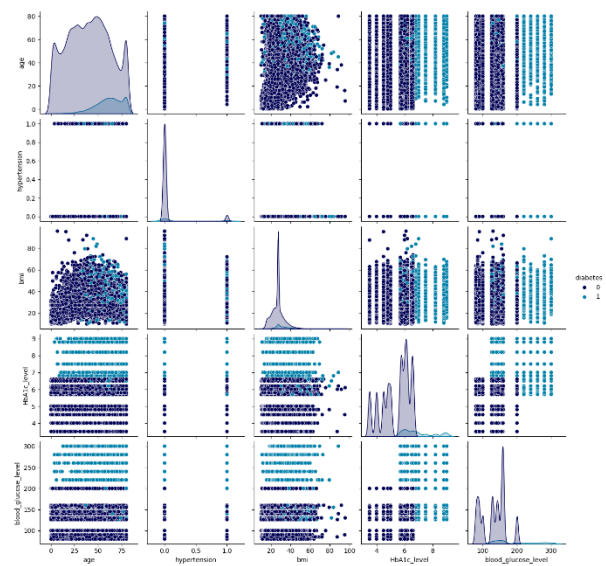
Studies using similar datasets for diabetes prediction have demonstrated that tree-based ensembles, like Random Forest and Gradient Boosting, often achieve high accuracy and AUC scores due to their effectiveness in handling mixed data types and class imbalance (Ganie, Pramanik, Malik, Mallik, & Qin, 2023).

## Research Questions

1. Which health and demographic indicators are most predictive of diabetes?
2. How do lifestyle factors (smoking, BMI) interact with clinical markers?
3. What is the best classification model for reliable diabetes prediction?

## Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted in an attempt of discovering existing relationships between the features in the dataset and the target variable (Diabetes). The pair plot of numerical features (Figure 20) reveals that diabetic individuals (teal) are concentrated at higher ages, BMI, HbA1c levels, and blood glucose levels compared to non-diabetic individuals (blue). Notably, HbA1c and blood glucose levels show strong class separation, indicating they are critical clinical markers. Additionally, those with hypertension = 1 have a higher likelihood of diabetes. Overall, age, obesity, poor glycemic control, and hypertension are key risk factors for diabetes in this dataset.



(Figure 20)

## Machine Learning – Classification

Data preprocessing was conducted (addressing unrealistic zero values) and encoding was done for categorical variables. Various classification models, including Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, CatBoost, AdaBoost, and K-Nearest Neighbors Classifier, were trained and tuned, followed by hyperparameter tuning for each model.

| Classification Model           | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|--------------------------------|----------|-----------|--------|----------|---------|
| Logistic Regression            | 0.96     | 0.95      | 0.96   | 0.95     | 0.96    |
| Decision Tree Classifier       | 0.97     | 0.97      | 0.97   | 0.97     | 0.97    |
| Random Forest Classifier       | 0.97     | 0.97      | 0.97   | 0.97     | 0.97    |
| XGBoost Classifier             | 0.97     | 0.97      | 0.97   | 0.97     | 0.98    |
| LightGBM Classifier            | 0.97     | 0.97      | 0.97   | 0.97     | 0.98    |
| CatBoost Classifier            | 0.97     | 0.97      | 0.97   | 0.97     | 0.98    |
| AdaBoost Classifier            | 0.97     | 0.97      | 0.97   | 0.97     | 0.97    |
| K-Nearest Neighbors Classifier | 0.96     | 0.96      | 0.96   | 0.95     | 0.93    |

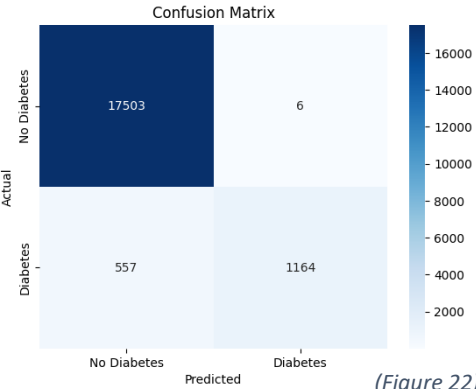
Classification in machine learning involves predicting the class of input data. To assess a classifier's true performance, evaluation metrics like **Accuracy, Precision, Recall, F1 score** and **ROC-AUC** are included (Rainio, Teuvo, & Klén, 2024)Accuracy indicates the proportion of correct predictions. Precision focuses on true positives among predicted positives, while recall identifies actual positives. The F1-score balances precision and recall, making it suitable for imbalanced datasets. The ROC curve compares recall to the false positive rate, and the AUC value reflects the model's ability to separate classes.

The **Random Forest Classifier** demonstrated strong performance with an **accuracy** of **0.97**. The model (Figure 22) identified 17,503 non-diabetic cases with only 6 false positives but missed 557 diabetic cases, resulting in a diabetes recall of 0.68 and a high precision of 0.99. The ROC curve (Figure 23) showed excellent discriminative ability with an **AUC** of **0.97**. The classification report (Figure 24) indicated an F1-score of 0.98 for non-diabetic cases and 0.81 for diabetic detection, yielding a high weighted average F1-score of 0.97.

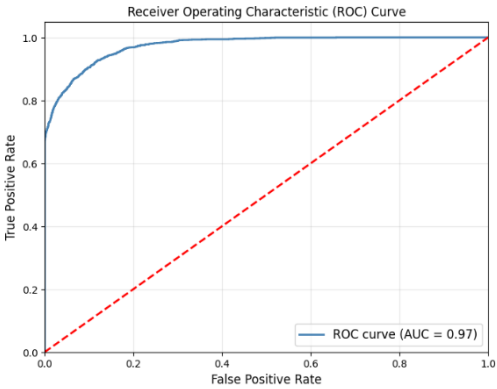
## Conclusion

| Classification Report: RandomForest Classifier |           |        |          |         |
|------------------------------------------------|-----------|--------|----------|---------|
|                                                | precision | recall | f1-score | support |
| No Diabetes                                    | 0.97      | 1.00   | 0.98     | 17509   |
| Diabetes                                       | 0.99      | 0.68   | 0.81     | 1721    |
| accuracy                                       |           |        | 0.97     | 19230   |
| macro avg                                      | 0.98      | 0.84   | 0.89     | 19230   |
| weighted avg                                   | 0.97      | 0.97   | 0.97     | 19230   |

(Figure 24)



(Figure 22)



(Figure 23)

The Random Forest Classifier achieved strong results in diabetes prediction with 97% accuracy and an AUC of 0.97. While it excelled at identifying non-diabetic cases and showed very high precision for diabetes, its lower recall (0.68) indicates missed detections of actual diabetic cases. This highlights the need for future improvements in recall, though overall the model remains a reliable tool for diabetes screening and risk assessment.

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